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Total No. of Questions : 5]

SEAT No. :

PA-2552

[Total No. of Pages : 4

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[5948]-102
M.C.A. (Management)
IT - 12 : DATA STRUCTURE AND ALGORITHMS
(2020 Pattern) (Semester - I)

Time : 2½ Hours] *[Max. Marks : 50*

Instructions to the candidates:

1) ***All questions are compulsory.***
2) ***From Q2 to Q5 having internal choices.***
3) ***Figure to be right indicate full marks.***

Q1) MCQ **[20×½=10]**

a) In a linked list insertion can be done as _____

i) beginning ii) end
iii) middle iv) all

b) The minimum number of fields with each node of DLL is

i) 1 ii) 2
iii) 3 iv) 4

c) In a stack if a user tries to remove an element from empty stack, it is called

i) empty collection ii) underflow
iii) overflow iv) garbage collection

d) Which method is used for retrieving the top element of the stack without deleting it

i) pop ( ) ii) dequeue ( )
iii) push ( ) iv) peek ( )

e) Binary tree is a special type of freedata structure in which every node can have a maximum _____ children

i) 4 ii) 2
iii) 1 iv) 0

f) A Binary search tree whose left subtree and right subtree differ in height by at most 1 unit is called

i) AVL tree ii) Red Black tree
iii) Lemma tree iv) Unique tree

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g) The Breadth first search traversal of a graph will result into?
   i) Linked List
   ii) Tree
   iii) Graph with back edges
   iv) None

h) Time complexity of DFS is (V-number of vertex, E-number of edges)
   i) O (V+E)
   ii) O(V)
   iii) O (E)
   iv) None

i) Heap can be used as _____
   i) Priority queue
   ii) stack
   iii) A decreasing order array
   iv) Normal array

j) What is the Best case for linearsearch
   i) O(nlog n)
   ii) O(logn)
   iii) O(n)
   iv) O(1)

k) In linear search with array, how many comparsions are needed in best case?
   i) 0
   ii) 1
   iii) n
   iv) n/2

l) Backtracking algorithm is implemented by constructing a tree of choice called as _____
   i) state - space tree
   ii) Back tracking tree
   iii) state - chart tree
   iv) Node tree

m) What is meant by the power set of a set?
   i) subset of all sets
   ii) set of all subsets
   iii) set of particular subset
   iv) emptyset

n) What is the other name of Dijkstra Algorithm?
   i) Single source shortest path
   ii) multiple source shortest path
   iii) multiple destination
   iv) single destination shortest path problem

o) Which of the following standard algorithms is not a Greedy algorithm.
   i) Dijkstra algorithm
   ii) Prim's
   iii) Kruskal
   iv) Bellment Ford stortest path

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p) What is the worst case complexity of quick sort?

i) O(n logn) ii) O(logn)

iii) O(n) iv) $O(n^2)$

q) Which of the following method is used for sorting in merge sort.

i) merging ii) partitioning

iii) selection iv) exchanging

r) In dynamic programming the output to stage n become the input to

i) stage n – 1 ii) stage n itself

iii) stage n + 1 iv) stage n – 2

s) Choose the recursive formula for Fibonacci series (n > = 1)

i) F(n) = F(n + 1) + F(n + 2) ii) F(n) = F(n) + F(n – 1)

iii) F(n) = F(n – 1) + F(n – 2) iv) F(n) = F(n – 1) – F(n – 2)

t) The algorithm like auick sort does not require extra memory for carrying out

i) in-place ii) stable

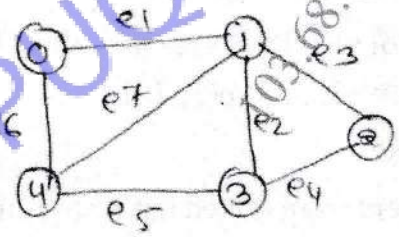
iii) unstable iv) in-partition

Q2) a) Draw Binary tree from given traversal **[5]**

inorder : 4, 2, 5, 1, 6, 7, 3, 8

Postorder : 4, 5, 2, 6, 7, 8, 3, 1

b) Give the Adjacency Matrix and linked list representation of undirected graph. **[3]**



c) Define collisions. **[2]**

OR

a) Apply the algorithm to draw Binary search tree for following data.

10, 08, 15, 12, 13, 07, 09, 17, 20, 18, 04, 05 **[5]**

b) Compare BFS and DFS. **[3]**

c) Explain Min Heap. **[2]**

Q3) a) Apply Rain Terrace algorithm to following problem
Input : [4, 2, 0, 3, 2, 5] Draw the figure and find solution. **[4]**

b) Write on algorithm for knight's Tour. **[3]**

c) Discuss use of Priority queve. **[3]**

OR

a) Apply the maximum subarray algorithm to the
Input : arr = [–2, –5, 6, –2, –3, 1, 5, –6] and find sum of maximum subarray. **[4]**

b) Give the explicit and implicit constraints in 8 queen's problem. **[3]**

c) Discuss Hamiltonian Cycle. **[3]**

Q4) a) Sort the following data using Mergesort [38, 27, 43, 3, 9, 82, 10]. **[4]**

b) Consider the following array [1, 3, 5, 8, 9, 2, 6, 7, 6] what is minimum number of jump required to reach the end of the array? **[4]**

c) Explain need of circular queve. **[2]**

OR

a) Write an algorithm of Kruskal's algorithm. **[4]**

b) Explain Rules for Tower of Hanoi with an suitable example. **[4]**

c) What is the purpose of Linked List. **[2]**

Q5) a) Write an algorithm to implement queue using linked list. **[3]**

b) Solve the following instance of 0/1 knapsack problem by applying Dynamic programming n = 3 w = (3, 5, 7) p(3, 7, 12), M = 4 **[7]**

OR

a) Write a algorithm to delete element from linked list whose sum is equal to zero. **[3]**

b) Find the largest common subsequence for the following string using Dynamic Programming **[7]**
X = [A, B, C, D, B, A, C, D, F]
Y = [C, B, A, F]

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